

Data-Driven Approximation of Earth–Mars Pork-Chop Surfaces

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Motivation & Problem Statement

- Interplanetary trajectory design requires repeated evaluation of transfer costs over large launch windows.
- These trade spaces are visualized using **pork-chop plots**:

$$\Delta V(\text{departure date, time-of-flight})$$

- Classical workflow:
 - Solve Lambert's problem at every grid point.
 - Becomes computationally expensive for wide windows or repeated optimization.
- **Project objective:**
 - Train a neural-network surrogate to approximate the Earth–Mars pork-chop surface.
 - Enable fast evaluation, grid search, and optimization using the surrogate.

Background & Related Work

- **RL for robust low-thrust trajectories:** Zavoli & Federici formulate Earth–Mars low-thrust transfers as Markov decision processes and train deep RL guidance and control networks that are robust to process noise, navigation uncertainty, and missed-thrust events.
- **Meta-RL for multiple uncertainties:** Federici & Zavoli extend this to meta-reinforcement learning with recurrent networks, learning a closed-loop policy that adapts across environments with mixed dynamic and control uncertainties for a fuel-optimal Earth–Mars transfer.
- **Surrogate NNs for trajectory planning:** Ruff et al. use neural-network surrogates inside a simulation-based optimization loop to design hypersonic approach-and-landing reference trajectories, showing large reductions in computation time versus direct simulation-based tuning.
- This project:
 - Adopts the idea of NN surrogates from Ruff et al., but in a simpler supervised setting.
 - Applies it to an impulsive Earth–Mars Lambert pork-chop rather than low-thrust optimal control.

Objective Function (What We Minimize)

We solve a two-impulse heliocentric transfer with Lambert's problem (Earth departure, Mars arrival).

Decision variables:

$$\mathbf{x} \triangleq \begin{bmatrix} t_{\text{dep}} \\ \text{TOF} \end{bmatrix}, \quad t_{\text{arr}} = t_{\text{dep}} + \text{TOF}$$

Lambert-based total cost:

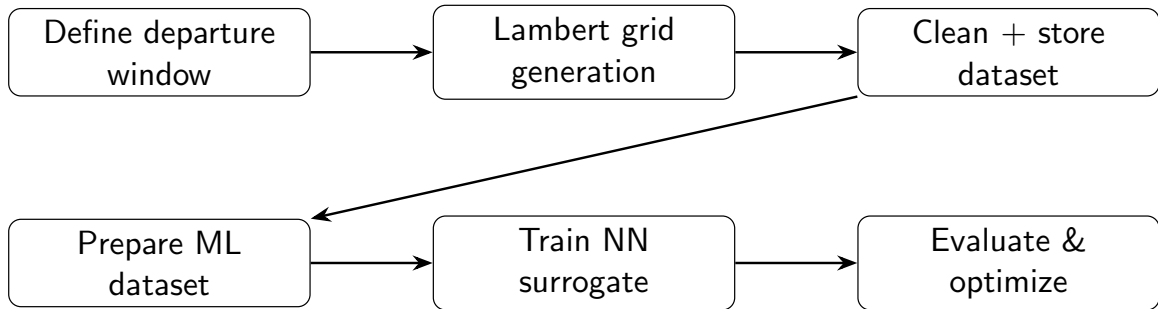
$$J(t_{\text{dep}}, \text{TOF}) = \Delta V_{\text{tot}} = \|\mathbf{v}_{\text{dep}}^* - \mathbf{v}_E(t_{\text{dep}})\| + \|\mathbf{v}_M(t_{\text{arr}}) - \mathbf{v}_{\text{arr}}^*\|$$

where $(\mathbf{v}_{\text{dep}}^*, \mathbf{v}_{\text{arr}}^*)$ are the Lambert transfer velocities solving

$$(\mathbf{v}_{\text{dep}}^*, \mathbf{v}_{\text{arr}}^*) = \text{Lambert}(\mathbf{r}_E(t_{\text{dep}}), \mathbf{r}_M(t_{\text{arr}}), \text{TOF}).$$

Feasibility / filtering: solutions with $\Delta V_{\text{tot}} > 20 \text{ km/s}$ are discarded (treated as invalid / NaN).

Workflow Overview



Methods & How We Compare

1) Ground-truth surface (Lambert grid):

- Sample t_{dep} daily over a 2-year window (~ 731 points).
- Sample TOF over 60 values within the derived feasible arrival bounds.
- Evaluate $J(t_{\text{dep}}, \text{TOF})$ at each grid point $\Rightarrow$ pork-chop surface.

2) “True objective” optimization (continuous):

- Optimize the same Lambert objective $J(t_{\text{dep}}, \text{TOF})$ over continuous bounds.
- Differential Evolution (DE) used to reduce sensitivity to nonconvexity / invalid regions.
- Output: best $(t_{\text{dep}}, \text{TOF})$ and optimization history (path over pork-chop).

3) Surrogate model (NN regression):

- Train $\hat{J}(t_{\text{dep}}, \text{TOF})$ on valid grid samples only.
- Evaluate: MAE/RMSE between $\hat{J}$ and J on held-out data.

4) Surrogate-based optimization (continuous):

- Minimize $\hat{J}(t_{\text{dep}}, \text{TOF})$ with DE.

Dataset Generation

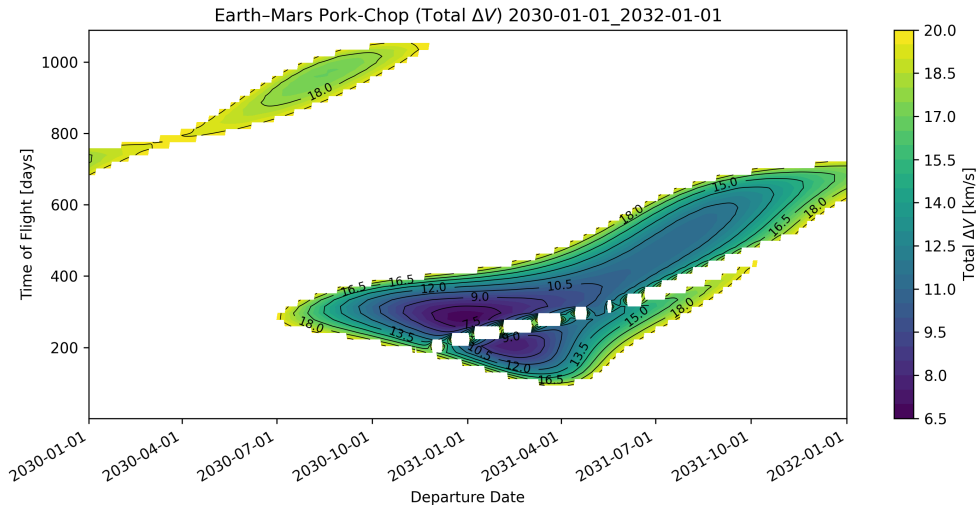
- Earth–Mars Lambert transfers using poliastro.
- Departure dates sampled daily over 2-year windows.
- Time-of-flight sampled uniformly within feasible arrival bounds (derived from arrival window).
- Invalid solutions ($\Delta V > 20$ km/s) removed.
- Multiple windows evaluated:

2030–2032, 2032–2034, 2034–2036, 2036–2038, 2038–2040

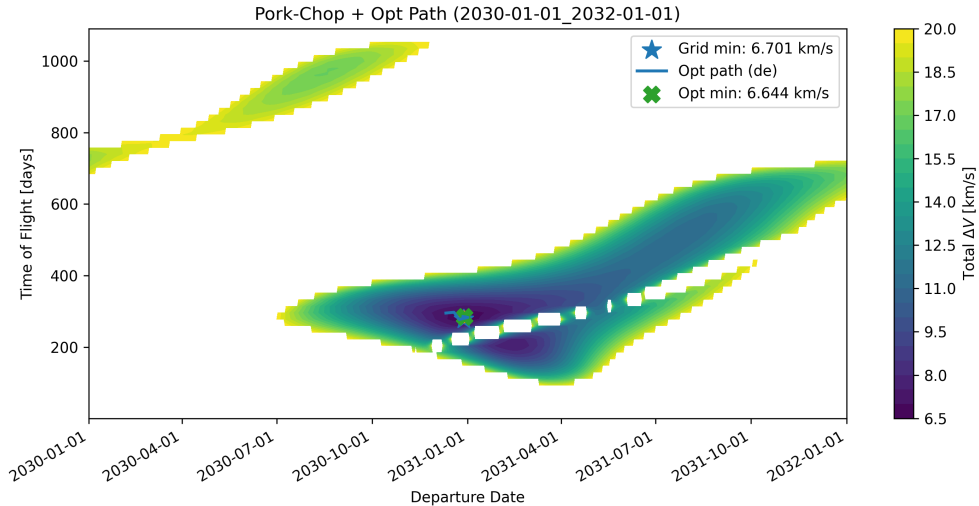
Run Summary (All Windows)

Window	Grid min	True opt min	Surrogate MAE	Surrogate RMSE
2030–2032	6.7009	6.6445	0.7313	1.2269
2032–2034	6.1420	6.1211	0.7985	1.3743
2034–2036	5.5770	5.5707	1.5362	3.0171
2036–2038	6.6032	6.6002	1.5831	3.1374
2038–2040	5.7429	5.7125	1.4953	2.5518

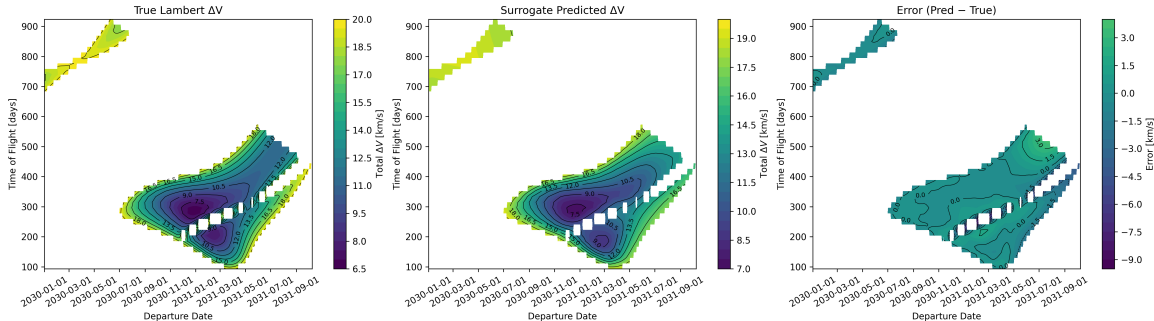
Baseline Pork-Chop Surface (Lambert)



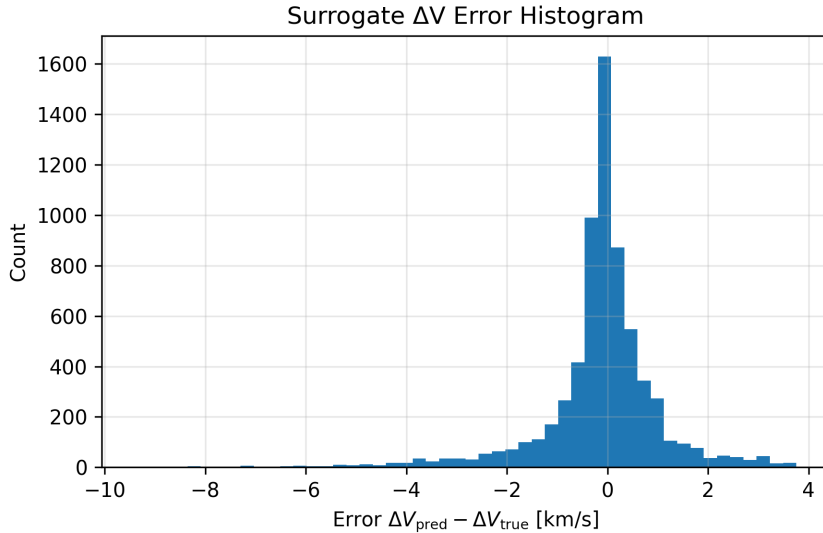
True Objective Optimization



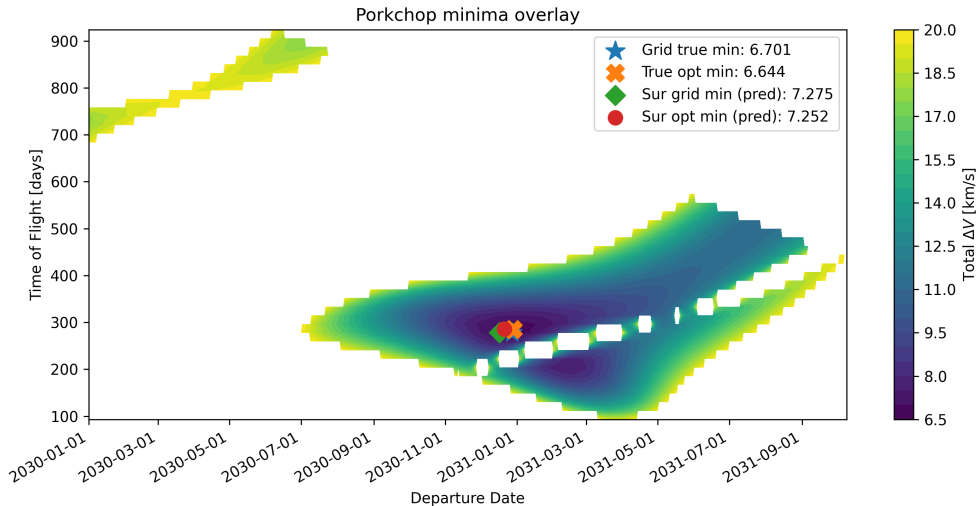
Surrogate Model Evaluation



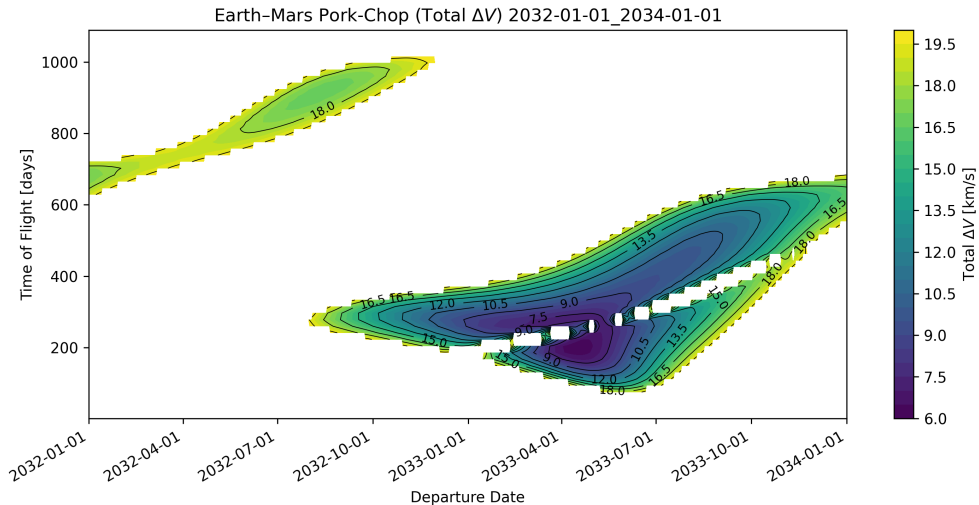
Error Distribution



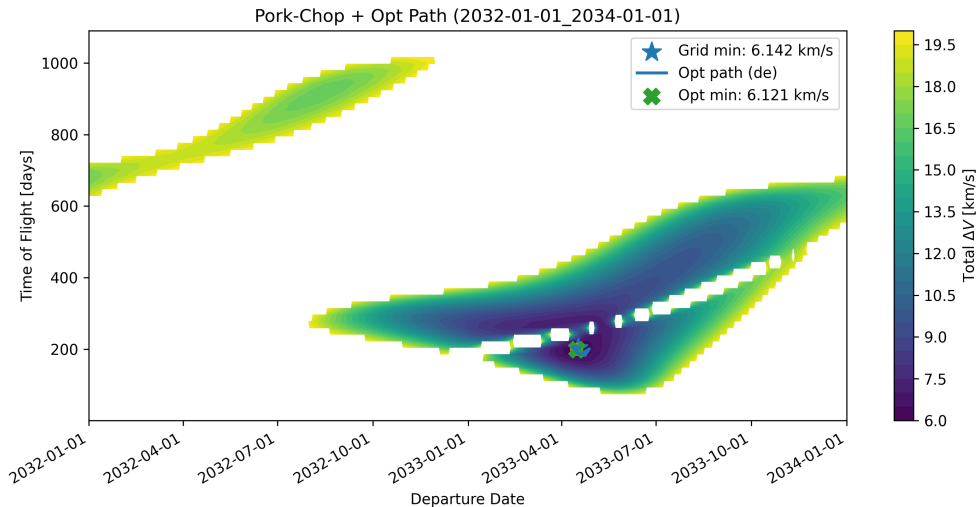
Four-Minima Comparison



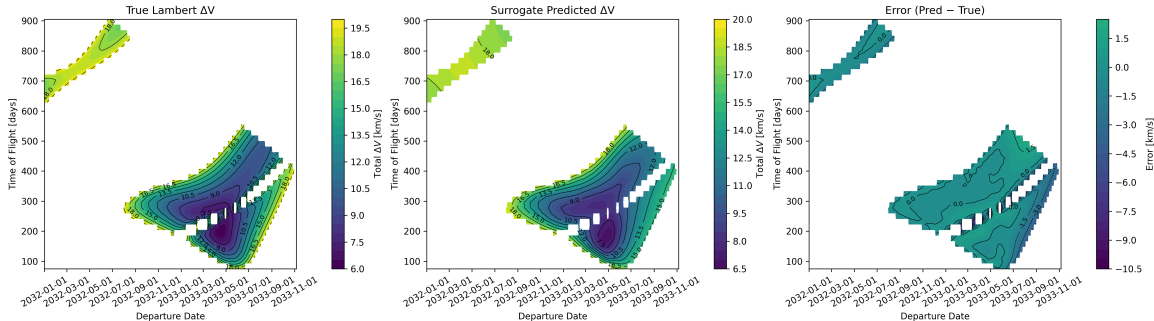
Backup: 2032–2034 Lambert Pork-Chop



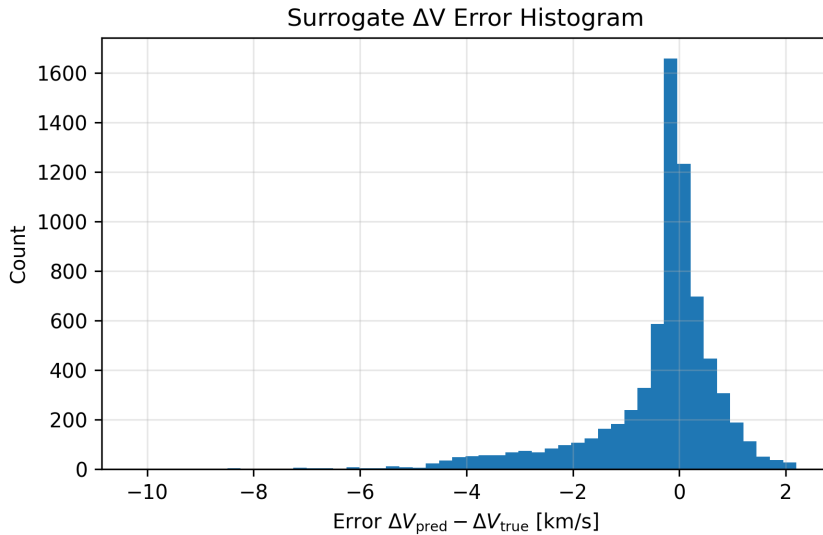
Backup: 2032–2034 True Objective Optimization



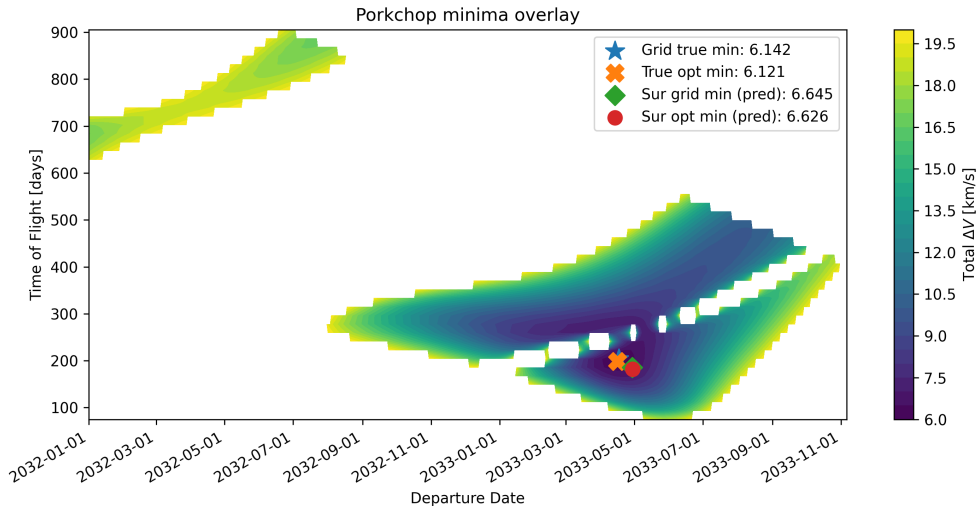
Backup: 2032–2034 Surrogate Triptych



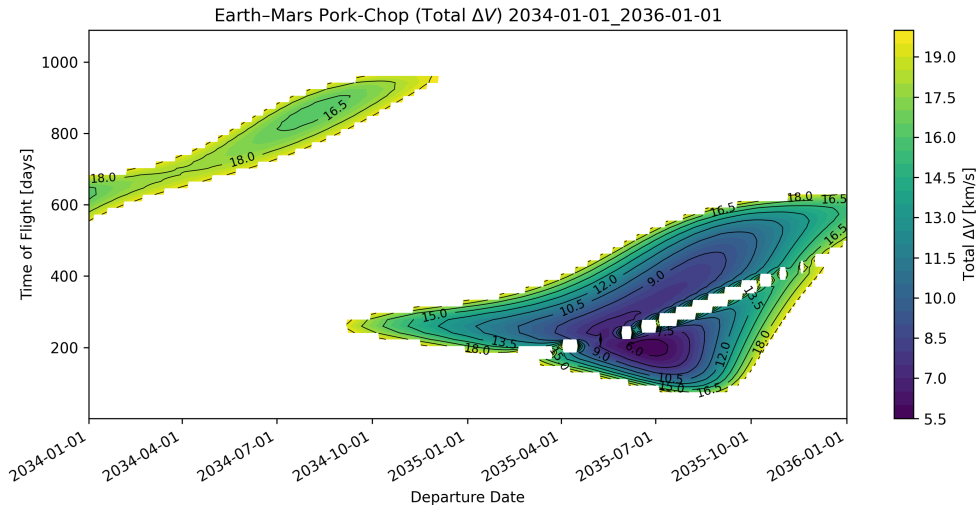
Backup: 2032–2034 Error Histogram



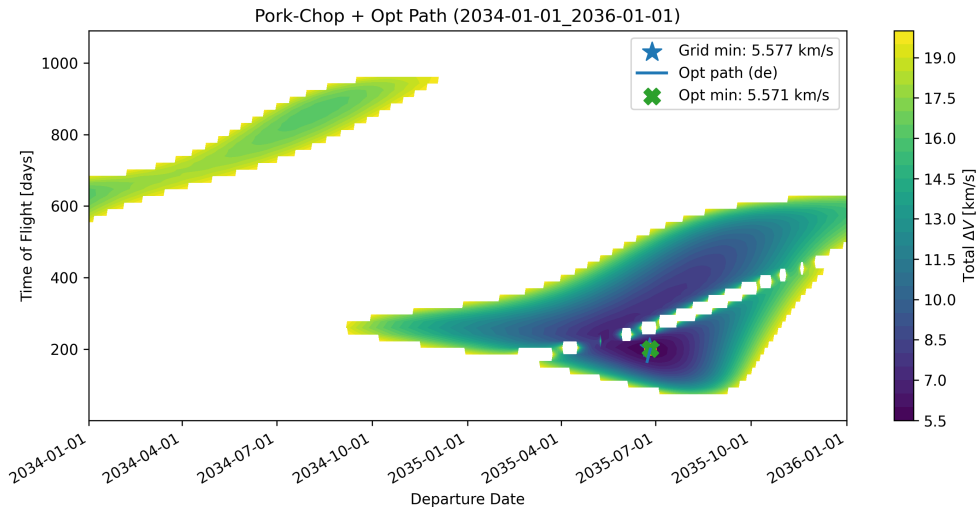
Backup: 2032–2034 Four-Minima Comparison



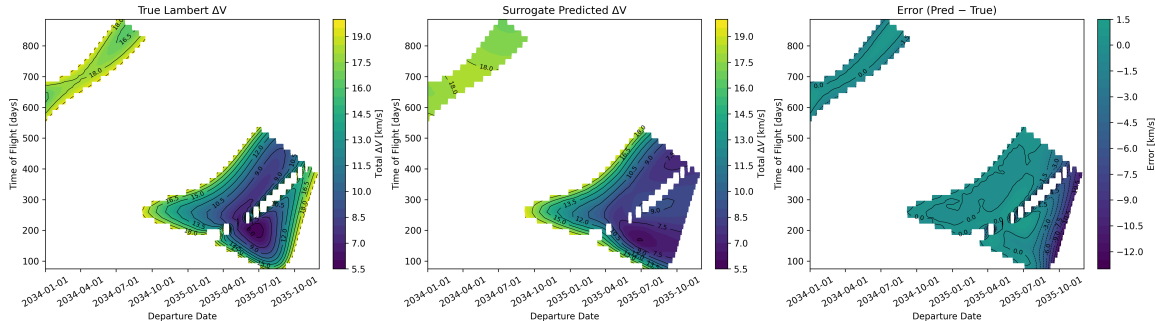
Backup: 2034–2036 Lambert Pork-Chop



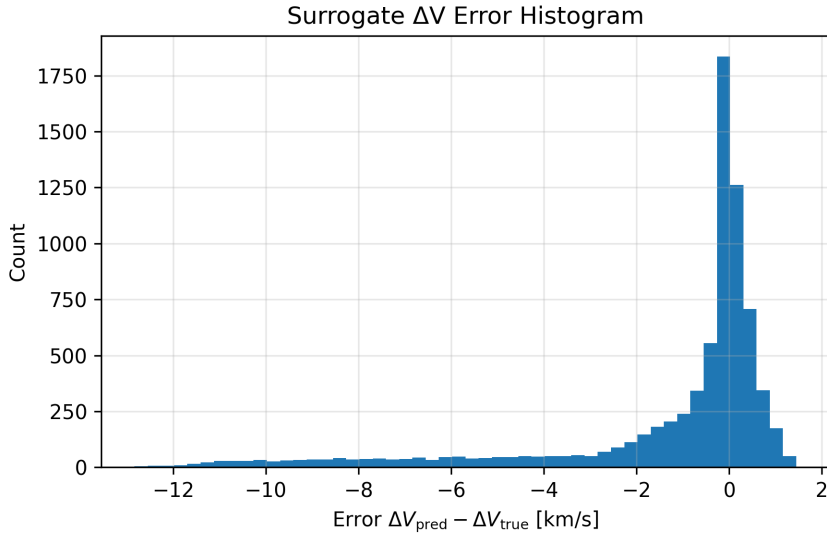
Backup: 2034–2036 True Objective Optimization



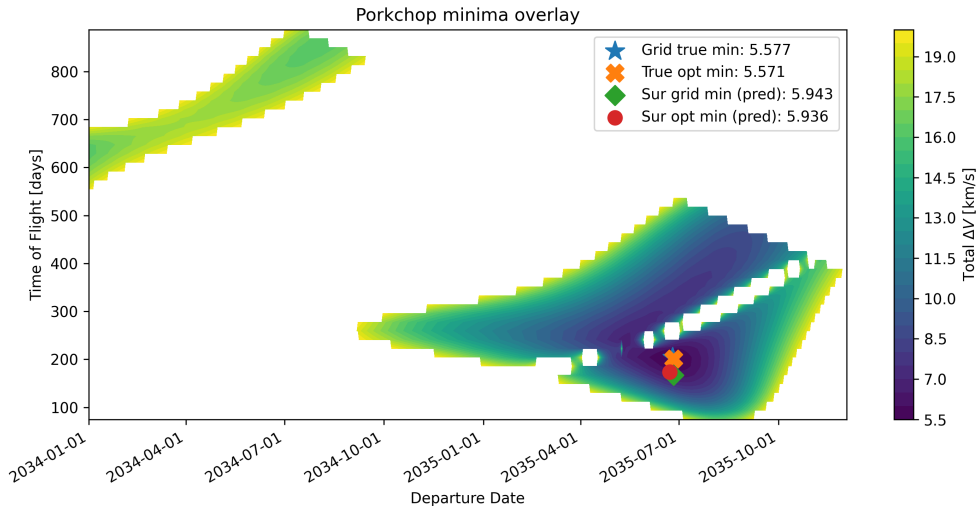
Backup: 2034–2036 Surrogate Triptych



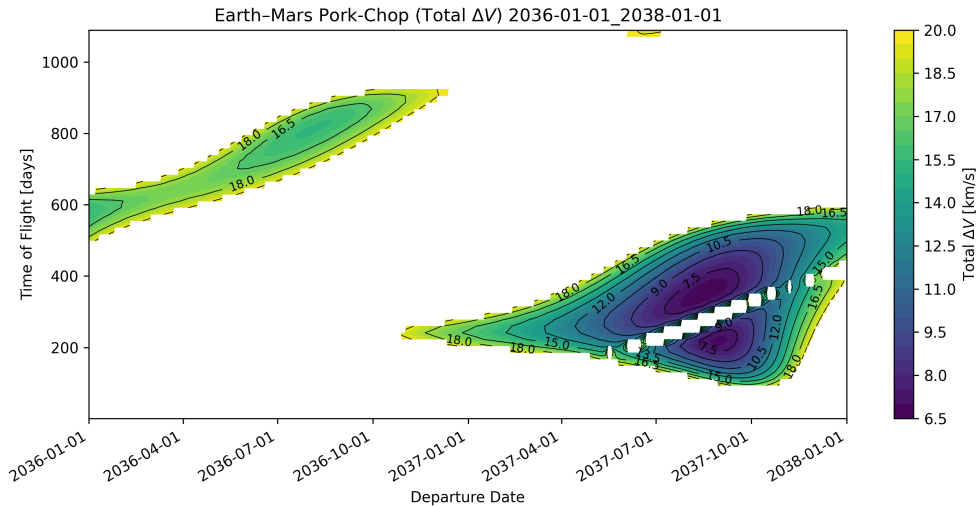
Backup: 2034–2036 Error Histogram



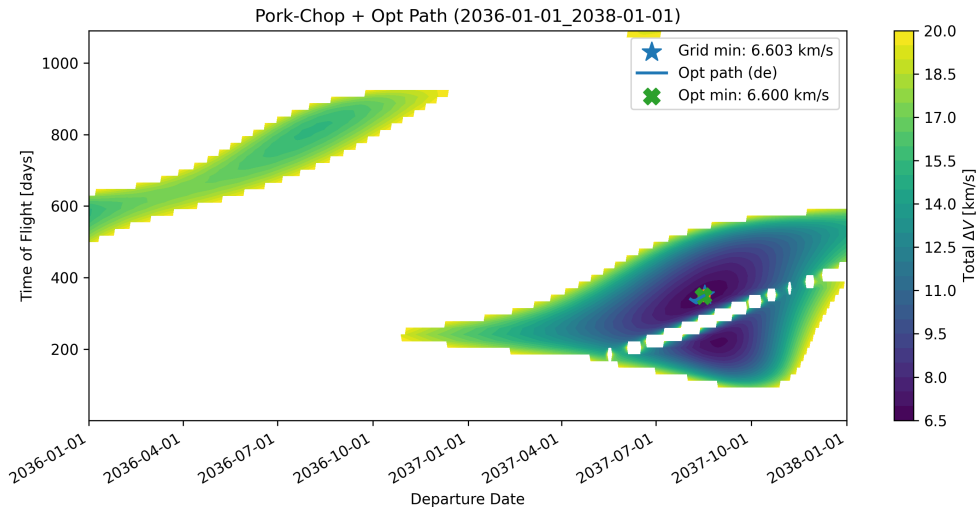
Backup: 2034–2036 Four-Minima Comparison



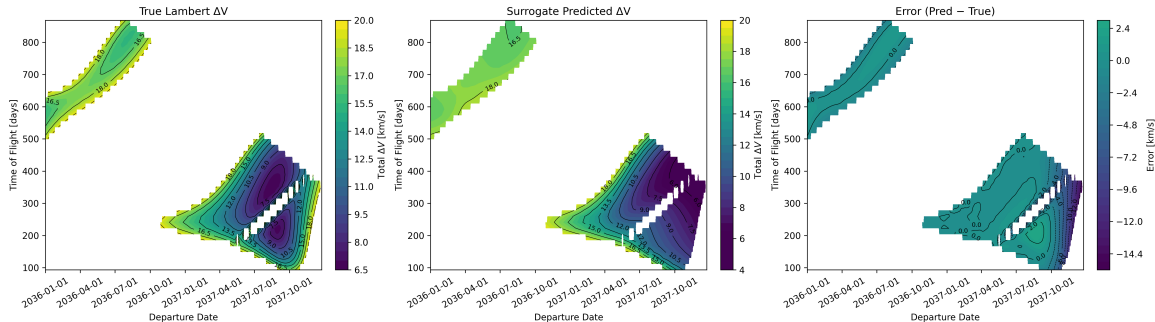
Backup: 2036–2038 Lambert Pork-Chop



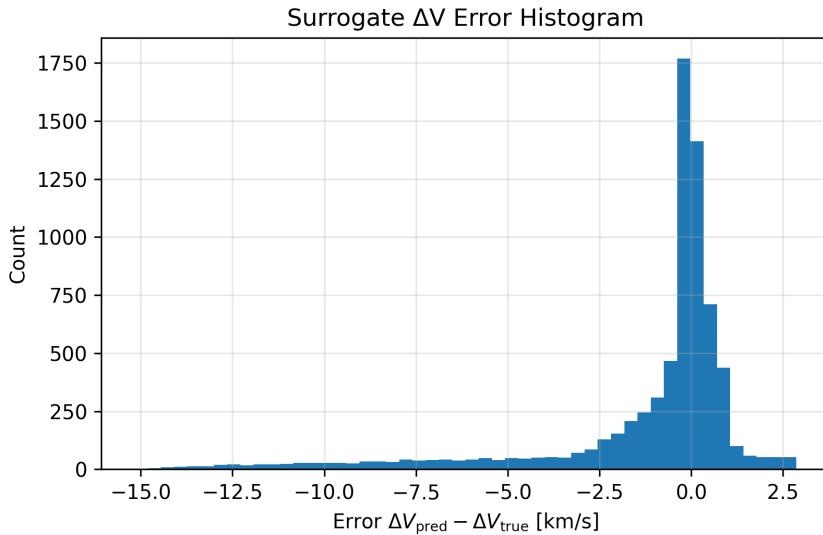
Backup: 2036–2038 True Objective Optimization



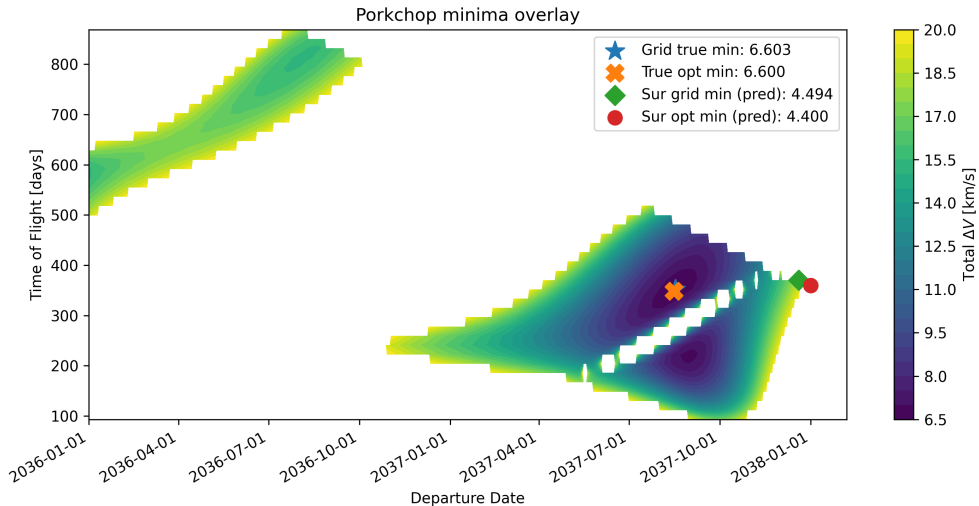
Backup: 2036–2038 Surrogate Triptych



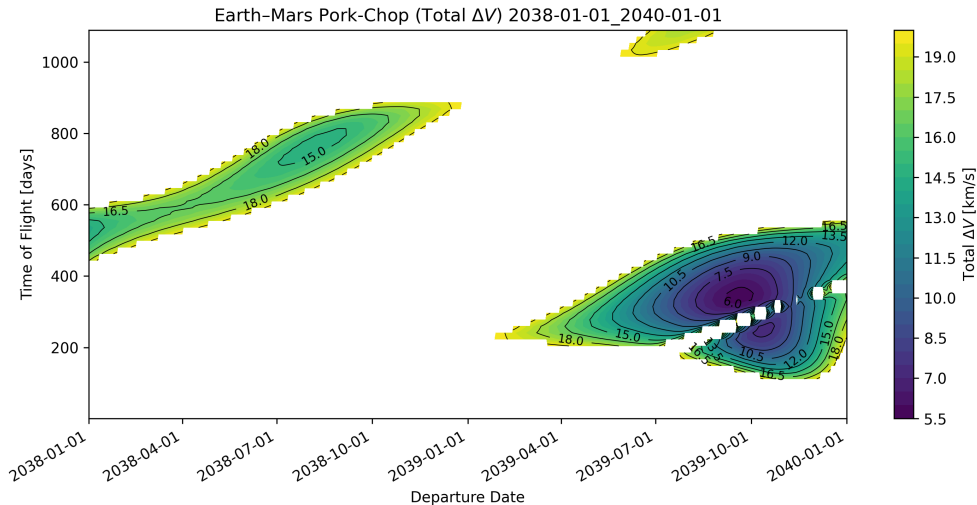
Backup: 2036–2038 Error Histogram



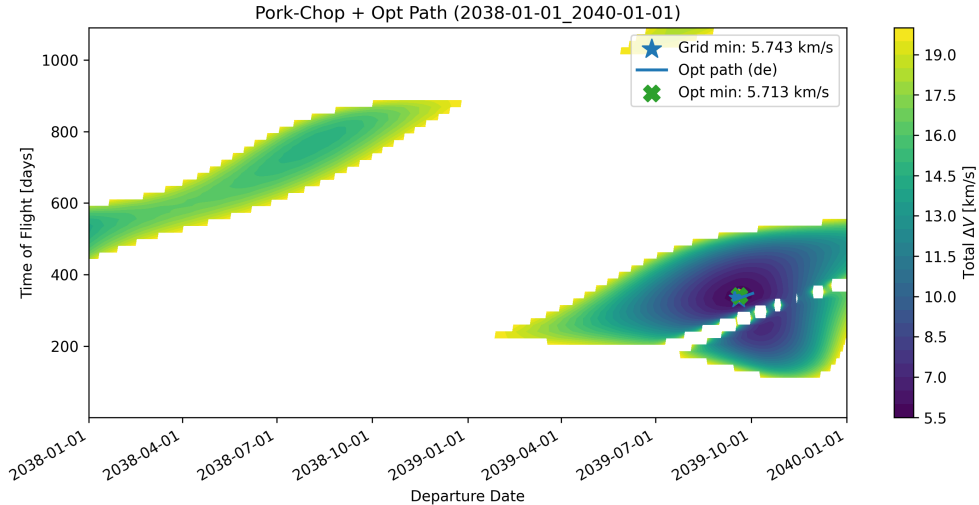
Backup: 2036–2038 Four-Minima Comparison



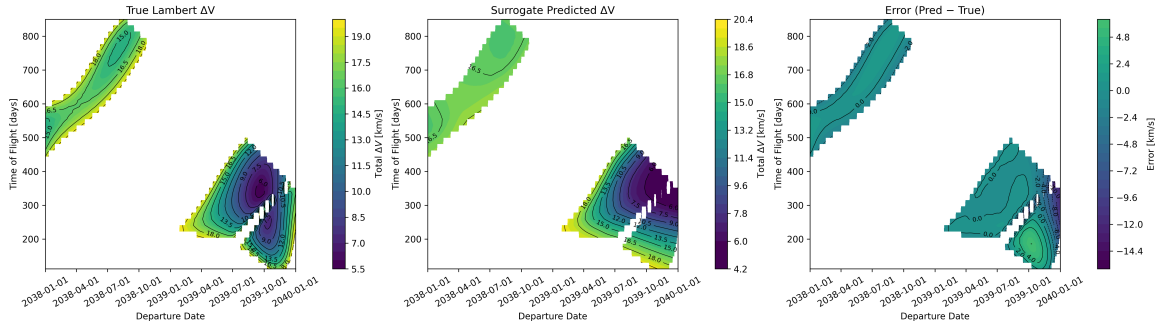
Backup: 2038–2040 Lambert Pork-Chop



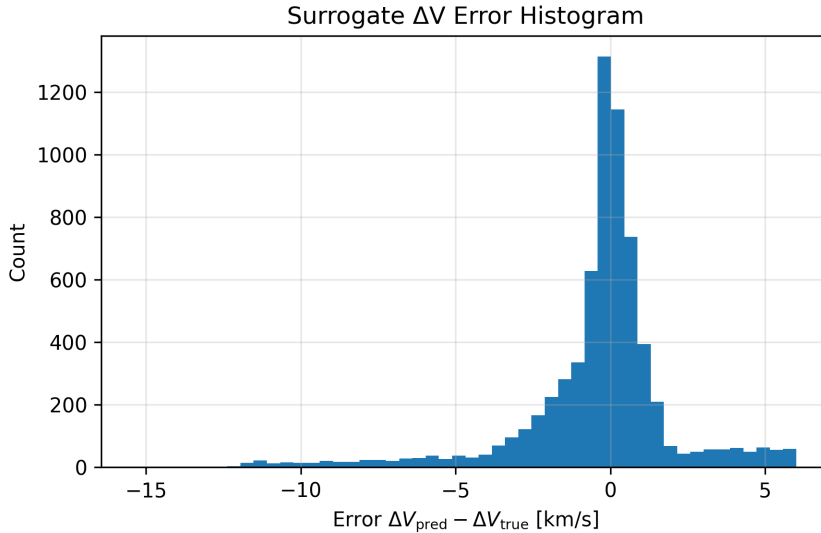
Backup: 2038–2040 True Objective Optimization



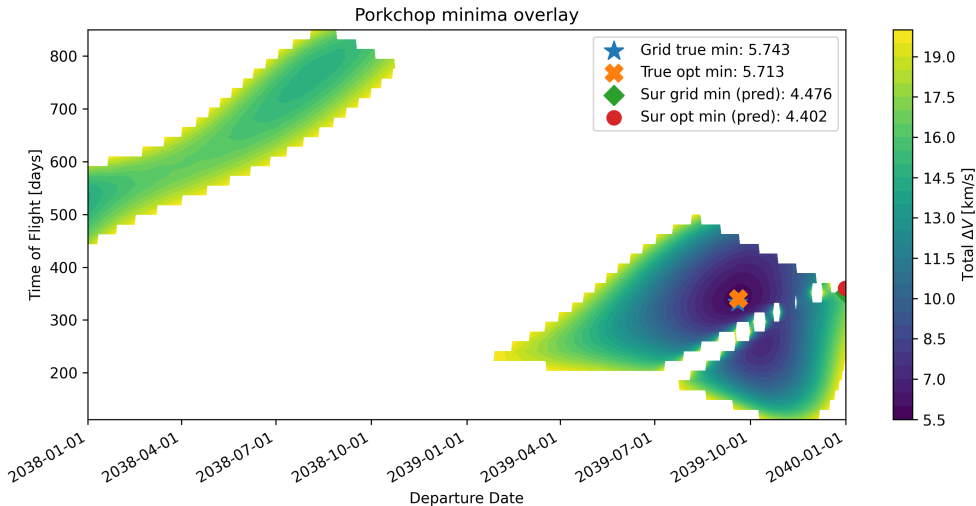
Backup: 2038–2040 Surrogate Triptych



Backup: 2038–2040 Error Histogram



Backup: 2038–2040 Four-Minima Comparison



References



A. Zavoli and L. Federici, "Reinforcement Learning for Low-Thrust Trajectory Design of Interplanetary Missions," AAS 20-616, preprint, 2020.



E. Ruff, R. Russell, M. Stoeckle, P. Miotto, and J. P. How, "Surrogate Neural Networks for Efficient Simulation-based Trajectory Planning Optimization," arXiv preprint arXiv:2303.17468, 2023.



L. Federici and A. Zavoli, "Robust Interplanetary Trajectory Design under Multiple Uncertainties via Meta-Reinforcement Learning," *Acta Astronautica*, 2024.

Questions?